Linear Equations & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

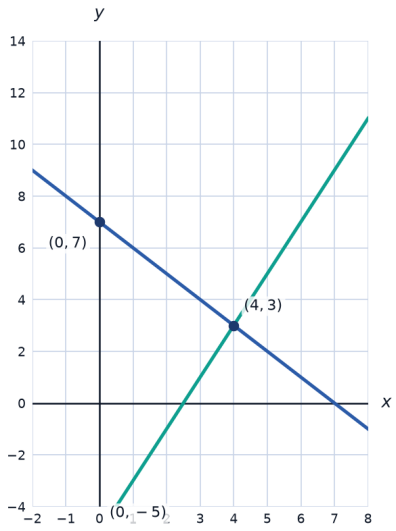
1 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the botanical conservatory, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



2 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

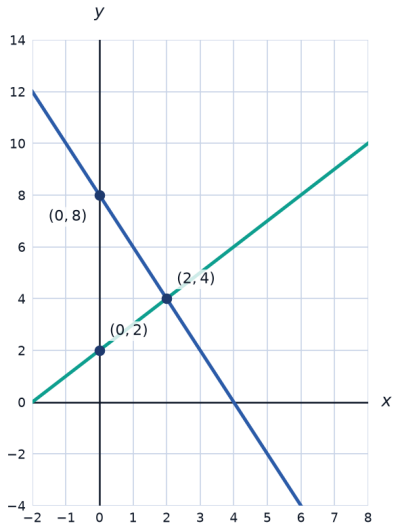
At the regional bakery, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



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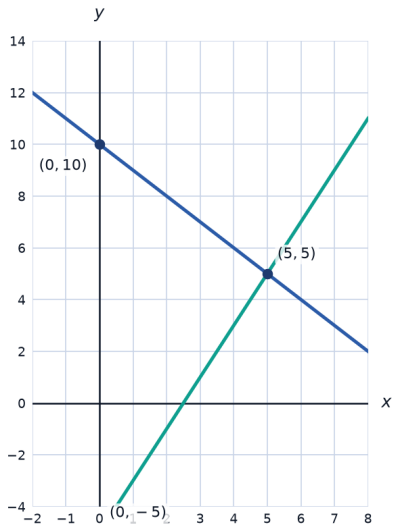
3 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the science museum, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



4 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

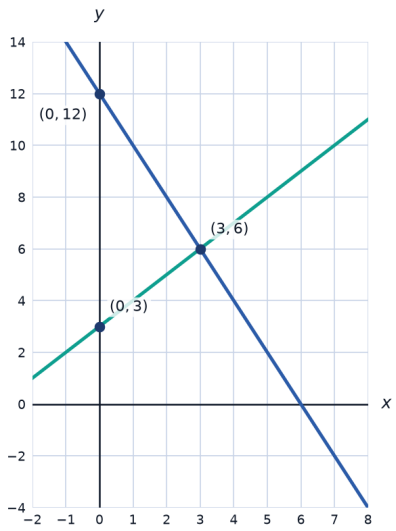
At the ceramic studio, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



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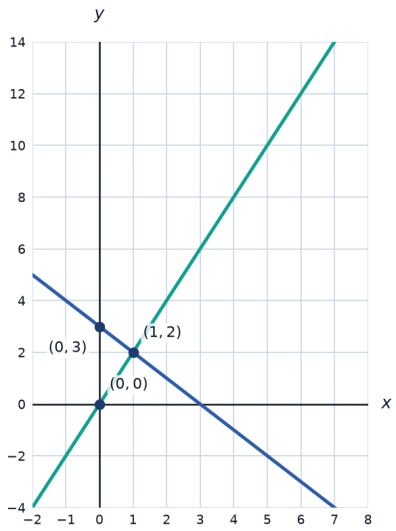
5 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the public transit depot, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



6 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

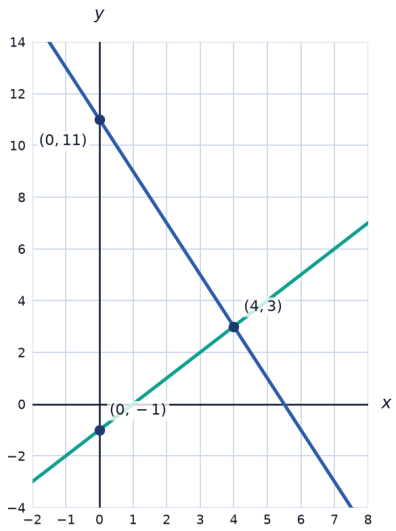
At the art gallery, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



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7 [Foundational] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the university lecture hall, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of x?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the orchestra hall, $\{6x + y = -3; 18x + ky = 7$ For what value of k does this system of equations have no solution?

9 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the honeybee apiary, a theater sold 29 tickets for a total of USD 310. Adult tickets cost USD 16 and child tickets cost USD 9. How many adult tickets were sold?

10 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the local farmers market, $\{6x - 3y = 0; 5x - 6y = 28$ What is the value of x?

11 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the performing arts academy, the real number t satisfies $\frac{23t+20}{2} = 391 + \frac{20}{2}$. What is the linear expression $p = 23t - 20$?

12 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the coastal aquarium, the linear system $17t + 6y = 284$ and $34t + (p + 6)y = 578$ has no solution. What is the constant parameter p?

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13 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the public transit depot, $\{3x + y = 2; 6x + ky = 14$ For what value of k does this system of equations have no solution?

14 [Intermediate] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the wildlife sanctuary, the ordered pair (t, y) satisfies $t + y = 26$ and $17t + 3y = 302$. What is the coordinate t in $p(t, y)$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the historic lighthouse, if the linear equation $6(x + 4) = 6x + c$ has infinitely many solutions for x, what is the value of the constant c?

16 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

At the botanical conservatory, adult tickets sell for USD 5 each and student tickets sell for USD 8 each. On Saturday, a total of 18 of these items were sold for a combined revenue of USD 114. How many adult tickets were sold on Saturday?

17 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

In the system of linear equations $3x + 4y = 11$ and $9x + ky = 26$ modeling two supply lines at the municipal library, k is a constant. If the system has no solution, what is the value of k?

18 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

For the community theater inventory model, the equation $\frac{1}{2}x + 4 = \frac{42}{6}$ relates the reserve count x to the target quota. What is the value of x?

19 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

For a calibration balance at the regional bakery, if the linear equation $6(x + 7) = 6x + c$ has infinitely many solutions for x, what is the value of the constant c?

20 [Challenging] • [SPR] • Target: Linear Equations in One Variable & Systems of Equations

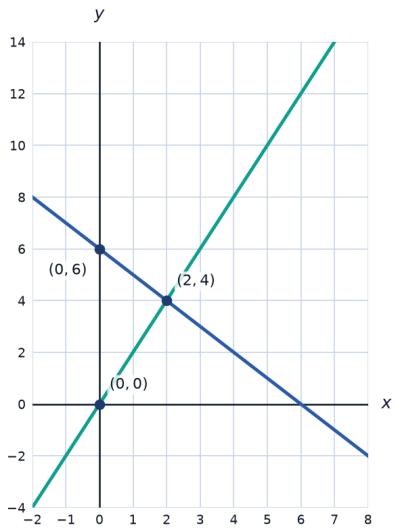
At the coastal aquarium, oak saplings sell for USD 9 each and maple saplings sell for USD 12 each. On Saturday, a total of 24 of these items were sold for a combined revenue of USD 246. How many oak saplings were sold on Saturday?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the municipal library, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y ?



a. 6

b. 3

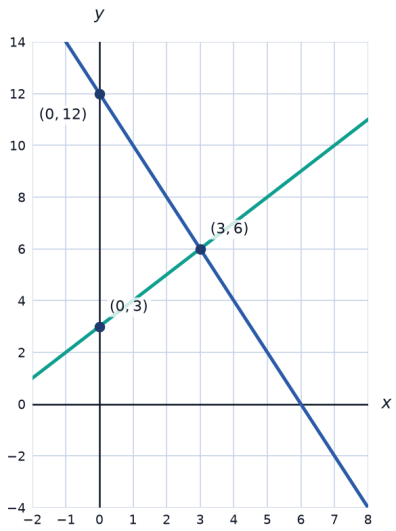
c. 9

d. 4

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22 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the community theater, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?



a. 8

b. 19

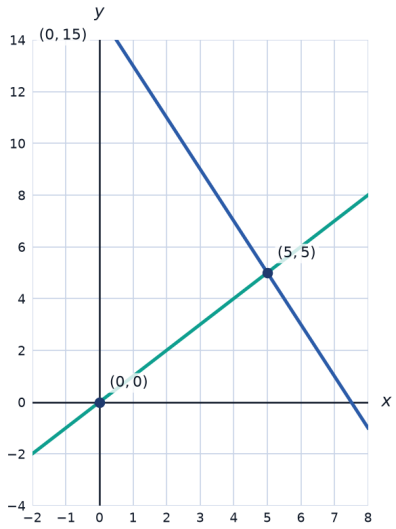
c. 9

d. 11

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23 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the coastal aquarium, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?



a. 5

b. 7

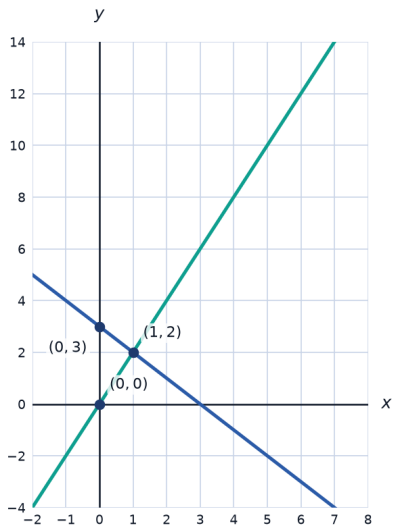
c. 4

d. 26

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24 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the mountain observatory, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?



a. 5

b. 2

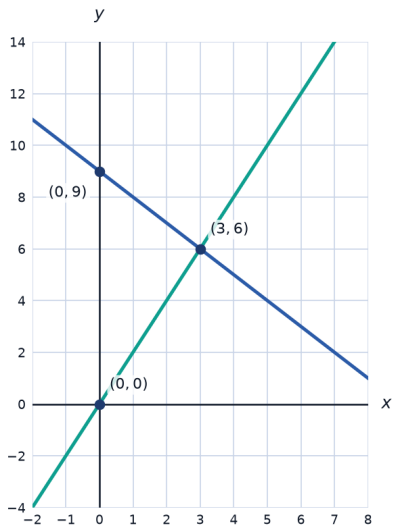
c. 6

d. 3

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25 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the urban bike-share station, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?



a. 19

b. 6

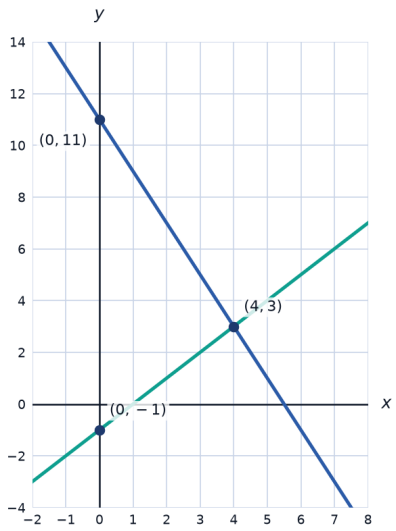
c. 8

d. 5

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26 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the solar microgrid facility, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?



a. 7

b. 9

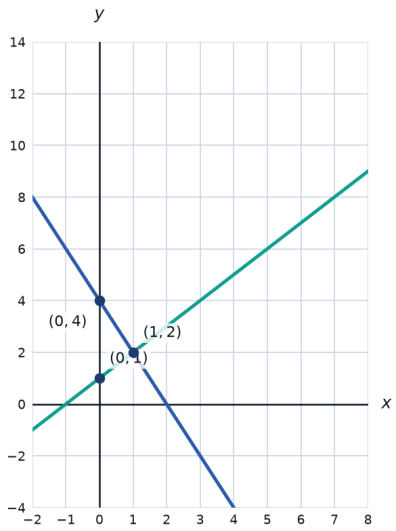
c. 6

d. 13

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27 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the agricultural cooperative, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of y?



a. 1

b. 3

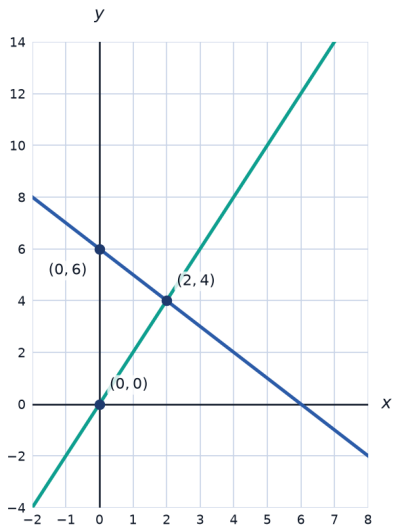
c. 2

d. 4

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28 [Foundational] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the publishing house, two linear cost constraints are graphed as lines ℓ_1 and ℓ_2 in the xy-plane shown. If (x, y) is the solution to the system of two linear equations represented by the graph, what is the value of $x + y$?



a. 9

b. 6

c. 8

d. 5

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the wildlife sanctuary, a store sells notebooks for USD 3 each and pens for USD 1.50 each. Maria spent USD 18 on a total of 8 items. How many notebooks did she buy?

a. 2

b. 3

c. 4

d. 5

30 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the charter ferry service, in the plane, the line $14x + 15y = 210$ intersects the axes at points A and B. What is the sum of intercepts f of $14x + 15y = 210$?

a. 29

b. 43

c. 31

d. 14

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31 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the textile workshop, in a calculation for the community orchard (case 44), what is the sum of all solutions to the equation $|2x - 10| = 4$?

a. 17

b. 10

c. 7

d. 16

32 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the geology lab, the ordered pair (x, y) satisfies $x + y = 16$ and $20x + 4y = 240$. What is the coordinate x in $f(x, y)$?

a. 7

b. 13

c. 31

d. 11

33 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the carpentry workshop, in the coordinate plane, consider the lines $f(x) = 6x - 14$ and $g(x) = 76$. What is the intersection coordinate x where $f(x) = g(x)$?

a. 21

b. 17

c. 15

d. 1

34 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the historic lighthouse, $\{6x + 6y = -8; 12x + ky = -7$ For what value of k does this system of equations have no solution?

a. 12

b. 14

c. 11

d. 21

35 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the community orchard, the point $A(19, 14)$ lies on the line $fx - 4y = 134$. What is the coefficient f?

a. 12

b. 10

c. 20

d. 6

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36 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the environmental club, a linear function $p(t)$ satisfies $p(0) = 9$ and $p(7) = 58$. What is the input t for which $p(t) = 100$?

a. 20

b. 15

c. 4

d. 13

37 [Intermediate] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the neighborhood hardware store, $\{6x - 6y = 54; 2x + 3y = -2$ What is the value of $x - y$?

a. 0

b. 9

c. 16

d. 11

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the honeybee apiary, in the system of linear equations $2x + 7y = 35$ and $6x + ky = 50$, k is a constant. If the system has no solution, what is the value of k?

a. 10

b. 6

c. 23

d. 21

39 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the carpentry workshop, for the textile workshop inventory model, the equation $\frac{1}{2}x + 7 = \frac{87}{6}$ relates the reserve count x to the target quota. What is the value of x?

a. 13

b. 22

c. 15

d. 17

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40 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the community orchard, at the geology lab, soccer balls sell for USD 9 each and training cones sell for USD 12 each. On Saturday, a total of 26 of these items were sold for a combined revenue of USD 270. How many soccer balls were sold on Saturday?

a. 14

b. 12

c. 16

d. 24

41 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In a model for the local farmers market, in the system of linear equations $6x + 4y = 39$ and $18x + ky = 54$, k is a constant. If the system has no solution, what is the value of k?

a. 7

b. 18

c. 14

d. 12

42 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

At the neighborhood bookstore, wooden chairs sell for USD 9 each and folding tables sell for USD 12 each. On Saturday, a total of 30 of these items were sold for a combined revenue of USD 318. How many wooden chairs were sold on Saturday?

a. 14

b. 16

c. 28

d. 17

43 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

In the system of linear equations $4x + 6y = 27$ and $12x + ky = 42$ modeling two supply lines at the robotics club, k is a constant. If the system has no solution, what is the value of k?

a. 9

b. 12

c. 20

d. 18

44 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For the university lecture hall inventory model, the equation $\frac{1}{2}x + 6 = \frac{57}{6}$ relates the reserve count x to the target quota. What is the value of x?

a. 5

b. 13

c. 7

d. 9

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45 [Challenging] • [MCQ] • Target: Linear Equations in One Variable & Systems of Equations

For a calibration balance at the culinary institute, if the linear equation $4(x + 5) = 4x + c$ has infinitely many solutions for x, what is the value of the constant c?

a. 24

b. 20

c. 9

d. 5

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